

$$q = 16, r = 0$$

$$\therefore A \rightarrow 4$$

$$q = -19, r = 5$$

$$\therefore C \rightarrow 3$$

$$q = -13, r = 1.$$

$$\therefore B \rightarrow 1$$

$$q = 14, r = 3$$

$$\therefore D \rightarrow 2$$

295. On dividing 534677 by 777 we get 101 as remainder.
 $\therefore (\text{Divisor}) - (\text{Remainder}) = (777 - 101) = 676.$

$$\begin{array}{r} 777 \overline{)534677} (688 \\ \underline{4662} \\ 6847 \\ \underline{6216} \\ 6317 \\ \underline{6216} \\ 101 \end{array}$$

296. $\frac{(\text{Dividend}) - (\text{Remainder})}{\text{Quotient}} = \text{Divisor}.$

$$\therefore \text{Divisor} = \frac{(940 - 25)}{15} = \frac{915}{15} = 61.$$

297. Remainder = 48, Divisor = $(5 \times 48) = 240.$

$$12 \times \text{Quotient} = \text{Divisor} \Rightarrow \text{Quotient} = \frac{240}{12} = 20.$$

$$\text{Dividend} = (240 \times 20) + 48 = 4848.$$

298. Quotient = 16, Divisor = $(25 \times 16) = 400.$

$$5 \times \text{Remainder} = \text{Divisor} \Rightarrow \text{Remainder} = \frac{400}{5} = 80.$$

$$\text{Dividend} = (400 \times 16) + 80 = 6480.$$

299. Divisor = $(555 + 445) = 1000$, Quotient = $2(555 - 445) = 2 \times 110 = 220$ and Remainder = 30.

$$\therefore \text{Required number} = (1000 \times 220) + 30 = 220000 + 30 = 220030.$$

300. Divisor taken = 12, Quotient obtained = 35, Remainder = 0.

$$\therefore \text{Dividend} = (12 \times 35) = 420.$$

$$\text{Now, dividend} = 420, \text{divisor} = 21, \text{remainder} = 0.$$

$$\therefore \text{Quotient} = \frac{420}{21} = 20.$$

301. Divisor = $7 \times \text{quotient} = 5 \times \text{remainder}$ and Dividend = $6 \times \text{remainder}.$

$$\begin{array}{r} 5x \overline{)6x(1} \\ \underline{5x} \\ x \end{array}$$

Let remainder be x . Then, divisor = $5x$ and dividend = $6x$.

On dividing $6x$ by $5x$, we get 1 as quotient and x as remainder.

$$\therefore \text{Quotient} = 1.$$

302. Since the required number is a 3-digit number, so on dividing by 19, it would yield a two-digit quotient which means that the quotient is greater than the remainder.

Let the remainder be x . Then, quotient = $x + 9$.

$$\text{So, number } N = 19(x + 9) + x = 20x + 171.$$

$$\therefore (N - 171) \text{ must be divisible by } 20.$$

$$\text{Clearly, } (371 - 171) = 200, \text{ which is divisible by } 20.$$

$$\text{Hence, the required number} = 371.$$

303. Let the number be x . Let x when divided by 136 give q as quotient and 36 as remainder. Then, $x = 136q + 36 = (17 \times 8q) + (17 \times 2) + 2 = 17 \times (8q + 2) + 2.$

So, the given number when divided by 17 gives 2 as remainder.

304. Let the number be x and the quotient be q .

$$\text{Then, } x = 195q + 47 = (15 \times 13q) + (15 \times 3) + 2 = 15(13q + 3) + 2.$$

So, the given number when divided by 15 gives 2 as remainder.

305. Let the number be x and the quotient be q .

$$\text{Then, } x = 899q + 63 = (29 \times 31q) + (29 \times 2) + 5 = 29(31q + 2) + 5.$$

So, the given number when divided by 29 gives 5 as remainder.

306. Let the number be x and on dividing by 5, we get q as quotient and 3 as remainder.

$$\begin{aligned} \text{Then, } x = 5q + 3 \Rightarrow x^2 &= (5q + 3)^2 = (25q^2 + 30q + 9) \\ &= 5(5q^2 + 6q + 1) + 4. \end{aligned}$$

Thus, on dividing x^2 by 5, we get 4 as remainder.

307. Let the smaller number be x . Then, larger number = $(x + 1365).$

$$\therefore x + 1365 = 6x + 15 \Rightarrow 5x = 1350 \Rightarrow x = 270.$$

Hence, the smaller number = 270.

308. Let $n = 4k + 3$.

$$\text{Then, } 2n = 2(4k + 3) = 8k + 6 = 4 \times 2k + 4 \times 1 + 2 = 4(2k + 1) + 2.$$

Thus, on dividing $2n$ by 4, we get 2 as remainder.

309. Let $x = 13p + 11$ and $x = 17q + 9$.

$$\text{Then, } 13p + 11 = 17q + 9 \Rightarrow 17q - 13p = 2 \Rightarrow q = \frac{2 + 13p}{17}$$

The least value of p for which $q = \frac{2 + 13p}{17}$ is a whole number, is $p = 26$.

$$\therefore x = (13 \times 26 + 11) = 338 + 11 = 349.$$

310. Let the dividend be $(x + 71)$ and the divisor be y .

$$\text{Then, } [2(x + 71) - 43] \text{ is divisible by } y \Rightarrow (2x + 142 - 43) \text{ is divisible by } y$$

$$\Rightarrow (2x + 99) \text{ is divisible by } y.$$

$$\therefore \text{Divisor} = 99$$

Shortcut Method:

$$\text{Divisor} = (2 \times 71 - 43) = (142 - 43) = 99.$$

311. Let $P = x + r_1$ and $Q = y + r_2$, where each of x and y are divisible by the common divisor.

$$\text{Then, } P + Q = (x + r_1) + (y + r_2) = (x + y) + (r_1 + r_2).$$

$(P + Q)$ leaves remainder r_3 when divided by the common divisor.

$$\Rightarrow [(x + y) + (r_1 + r_2) - r_3] \text{ is divisible by the common divisor.}$$

Since $(x + y)$ is divisible by the common divisor, so divisor = $r_1 + r_2 - r_3$.

312. As proved in the above question, divisor = $4375 + 2986 - 2361 = 5000$.

313. The number is of the form $(13k + 1)$, where k is of the form $(5m + 3)$.

$$\therefore \text{Number} = 13k + 1 = 13(5m + 3) + 1 = 65m + 40.$$

Clearly when the number is divided by 65, we get 40 as remainder.

314. Clearly, $(2272 - 875) = 1397$, is exactly divisible by N .

$$\text{Now, } 1397 = 11 \times 127.$$

$\therefore$ The required 3-digit number is 127, the sum of whose digits is 10.

315. $z = 13 \times 1 + 8 = 21.$

$$\begin{array}{r} 9 \mid x \\ 11 \mid y - 8 \\ 13 \mid z - 9 \\ \hline 1 - 8 \end{array}$$

$$y = 11 \times z + 9 = 11 \times 21 + 9 = 240.$$

$$x = 9 \times y + 8 = 9 \times 240 + 8 = 2168.$$